

Scored Against History: A Pilot Study of Hindsight Leakage and Strategic Learning in an LLM-Simulated Organization

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<https://github.com/LoopGlitch26/Frontier-Autolab>

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Abstract

History is an attractive answer key for evaluating strategic reasoning: a decision made with information available in 1996 can be scored against what actually happened afterwards. Large language models (LLMs) make this answer key leaky, because they already know the history. We introduce FRONTIER AUTOLAB, an open protocol in which a simulated startup, voiced by sixteen LLM role personas, must reinvent itself across nine technology eras from 1990 to 2040. In six *training* eras (1990–2020) the organization decides using only dated information; an LLM historian-judge then reveals what happened, scores the call on five dimensions, and appends lessons to a Playbook that carries forward. In one live era (September 2026) and two forecast eras (2032, 2040) there is no answer key. We report four exploratory trajectories (36 era decisions, 180 subscores) produced under three orchestration schemes. In all four, training-era scores rise across eras (+1.4 to +2.6 points per era) while the judge’s own hindsight-leakage subscore worsens in the same eras (per-run Pearson r between total and hindsight subscore from -0.38 to -0.92 , $n = 6$ each), so apparent learning cannot be separated from recall. Layer choice is the lowest-scoring dimension in every run, consistent with a qualitative pattern in which the organization named the coming capability shift but chose an adjacent, asset-reachable layer. We also document measurement failures that any “learn from history” agent benchmark should expect: inconsistent score aggregation, a column-shift error in exported scores, probable cross-run contamination, briefing-selection leakage, and a rubric that rewarded one run for fifty years of gated, never-built pilots. We release all prompts, run records, scores and an API harness with ablations, and we specify a controlled follow-up design with pre-registered hypotheses.

1 Introduction

How well can an LLM-based organization reason about technological change? Existing multi-agent systems show that role-structured LLM agents can collaborate on software, simulation and open-ended tasks [2, 11, 13, 19, 24, 25], and LLM forecasting systems now approach human crowds on some question sets [12, 17]. Strategy, however, is harder to evaluate than code or calibrated probabilities. The quality of a strategic call is usually revealed years later, if at all.

History offers an answer key. If an organization decides in January 1996 using only what was public then, we can compare its decision with what happened from 1996 to 2001. This is the logic of backtesting in finance and of retrospective forecasting benchmarks. For LLMs the key is leaky: a model trained on text written after 1996 knows which companies won. The problem is well known for question-answering benchmarks and forecasting evaluations [3, 9, 10, 22, 27]. It is sharper for open-ended strategy, where the model’s own reasoning chooses what to mention and an LLM judge decides what counts as foresight.

This paper presents FRONTIER AUTOLAB, a protocol and set of artifacts for studying that problem in a setting that is deliberately open-ended. One simulated company, voiced by sixteen named role personas and two external judges, must reinvent itself in each of nine eras. The judge scores every decision, and the organization carries lessons forward in a Playbook, so we can ask whether scores improve over eras and why.

We report four exploratory trajectories. They are not a controlled experiment, and we are explicit about why. Their value is in what they reveal about measurement:

- **Learning and leakage move together.** In every run, training-era scores rise while the hindsight subscore worsens in the same eras (Section 5.2).
- **A consistent weakness in layer choice.** Across all runs, the judge rates layer choice lowest, matching a qualitative pattern of “naming the frontier, then building the adjacent layer” (Section 5.3).
- **Rubrics can reward paralysis.** One run scored highest while never building a product in fifty simulated years (Section 5.4).
- **Mundane measurement failures matter.** Inconsistent aggregation, an export error and probable contamination between runs each change what the data appear to say (Section 5.6).

Contributions. (1) An open protocol for scoring agent-organization strategy against dated history, with role charter, per-era loop, rubrics and anti-hindsight rules. (2) Four fully published trajectories (36 era decisions) including every briefing, memo, board decision and reveal where available, one disclosed operator intervention and all redactions. (3) An analysis of hindsight leakage and related measurement failures as confounds for any “learn from history” agent benchmark. (4) A controlled follow-up design with pre-registered hypotheses, ablations, leakage controls and judge-validity checks, supported by an open API harness.

2 Related work

LLM agent societies. Role-playing and multi-agent frameworks assign LLM instances roles and communication protocols to solve tasks or simulate social behaviour [2, 13, 19, 24, 25]; see Guo et al. [11] for a survey. Multi-agent debate can improve factuality and diversity of reasoning [6, 20]. Agents that accumulate verbal lessons or skill libraries across episodes [29, 33] are the closest analogue to our Playbook. We differ in evaluating a sequence of strategic decisions against real historical outcomes rather than task success.

LLMs as simulated humans and organizations. LLMs have been used as simulated economic agents [14] and survey respondents [1]. Our organization is a simulated firm; like these works we treat its outputs as model behaviour, not as evidence about real firms.

LLM-as-a-judge. LLM judges agree with human preferences on many tasks but show position and verbosity biases [34, 35] and favour their own generations [23]. In all our runs the judge shares a model family with the players, and in one run the same context wrote the decisions it judged.

Table 1: Eras. The E6 window runs to the live era and is longer than the others.

Era	Start	Mode	Scored against
E1	Jan 1990	training	history, 1990–1995
E2	Jan 1996	training	history, 1996–2001
E3	Jan 2002	training	history, 2002–2007
E4	Jan 2008	training	history, 2008–2013
E5	Jan 2014	training	history, 2014–2019
E6	Jan 2020	training	history, 2020–Aug 2026
E7	Sep 2026	live	current market, checked with web search
E8	Jan 2032	forecast	projected world (no ground truth)
E9	Jan 2040	forecast	projected world (no ground truth)

Contamination, cutoffs and temporal leakage. Benchmark contamination can inflate scores [10, 27], effective knowledge cutoffs differ from reported ones [3], and LLM-based backtests suffer look-ahead bias [9]. Paleka et al. [22] catalogue pitfalls in evaluating LLM forecasters, including temporal leakage in retrospective evaluation. Our setting adds leakage through the choice of what a dated briefing mentions and through a judge that labels foresight after the fact.

Forecasting benchmarks. Autocast [36], retrieval-augmented forecasting systems [12] and ForecastBench [17] evaluate probabilistic forecasts of resolvable questions, following the tradition of Tetlock [31], Tetlock and Gardner [32]. Strategy decisions are not resolvable questions, which is why we rely on a judge and why judge validity matters.

Organizational learning and strategy. The protocol operationalizes ideas from organizational learning [18], the exploration–exploitation trade-off [21], absorptive capacity [5], dynamic capabilities [30] and disruptive innovation [4]. Its Red Team draws on devil’s-advocacy research [28] and the risk of groupthink [15]. Hindsight bias in human judgment [7, 26] is the human analogue of the leakage we study.

3 The Frontier Autolab protocol

3.1 Eras

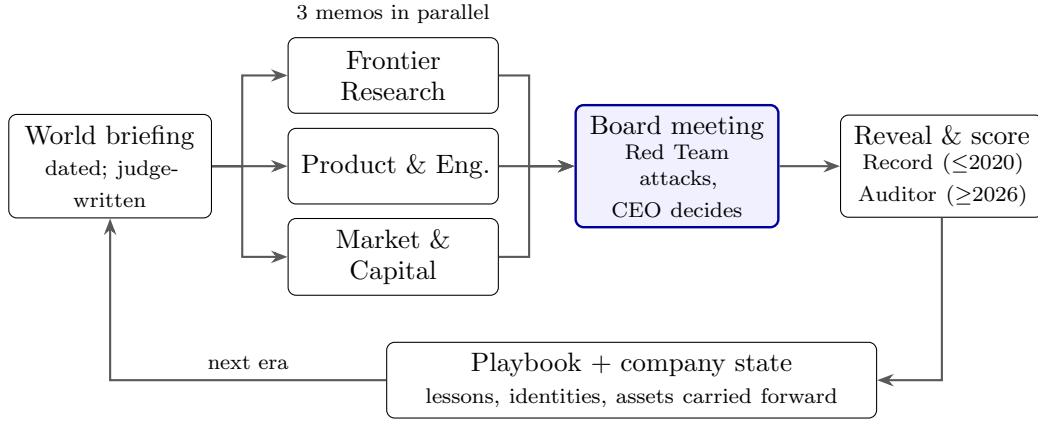
A single company is carried through nine eras (Table 1). Its only mandate is *reinvention toward the frontier*: in each era it should become the most important company that era could produce, even if that means ending its current business. Training eras are scored against real history over a roughly six-year window. The live era is scored by an auditor who checks claims against the current market with web search. Forecast eras have no answer key.

3.2 Roles

The charter defines sixteen named personas in five groups and two external judges (Table 2). A persona is a role with a lens, voiced by model calls. How personas map to model contexts differs by run (Section 4); in no run was each persona a separately instantiated autonomous agent.

Table 2: Role charter.

Group	Personas	Lens
Executive	4	CEO (final call), CTO (feasibility), Chief Scientist (which curves bend), CSO (where value pools)
Frontier Research	4	Emerging capabilities, historical analogies, scenarios, what cannot yet be measured
Product & Engineering	4	What a small team can ship, the wedge, unavoidable infrastructure, ecosystems
Market & Capital	3	Who pays first, financing climate, cost curves and margins
Governance	1	Red Team: attacks every proposal, hunts hype and hindsight
External judges	2	The Record (training eras), The Auditor (live and forecast eras)

**Figure 1:** The per-era loop. From E2 on, memos also read the previous era’s board decision and reveal. The judge that scores an era also writes the next era’s briefing.

3.3 The per-era loop

Figure 1 and Algorithm 1 summarize one era. The judge writes a dated world briefing. Three departments write memos in parallel, each proposing two theses as the chain *capability* $\rightarrow$ *adoption* $\rightarrow$ *bottleneck* $\rightarrow$ *layer owned* $\rightarrow$ *data* $\rightarrow$ *next capability*. The board meets, the Red Team attacks, and the CEO decides the company’s identity, thesis, what it kills and keeps, its wedge product, kill criteria and a dissent log. The judge then reveals what happened (training) or audits the claims (live and forecast), scores the call, and appends lessons to the Playbook. The judge also writes the next era’s briefing.

3.4 Rules

Five rules apply throughout. *Anti-hindsight*: in training eras, memos and decisions may use only information public before the era’s start date, enforced by the Red Team and penalized by the judge. *Reinvention by default*: keeping the previous business must be argued for. *Layer thinking*: every thesis states the full chain above. *Small-team realism*: the wedge must be buildable by fewer than twenty people with era-realistic capital. *Playbook primacy*: from E2, every memo cites at least one Playbook lesson or argues against it.

Algorithm 1 One FRONTIER AUTOLAB trajectory

Require: charter C , eras $E_1, \dots, E_9$, player model π , judge model J

```
1:  $S \leftarrow$  initial company state ($1.5M seed, 16 personas);  $P \leftarrow \emptyset$  ▷ Playbook
2:  $B_1 \leftarrow J.BRIEFING(E_1)$  ▷ only facts dated before  $E_1$ 
3: for  $k = 1$  to 9 do
4:   for  $d \in \{\text{frontier, product, market}\}$  in parallel do
5:      $M_d \leftarrow \pi.MEMO(d, C, S, P, B_k, D_{k-1}, R_{k-1})$  ▷ two theses each
6:   end for
7:    $(D_k, S) \leftarrow \pi.BOARD(C, S, P, B_k, \{M_d\}, \text{red\_team} = \text{on})$ 
8:   if  $E_k$  is a training era then
9:      $(R_k, \mathbf{s}_k, L_k) \leftarrow J.REVEAL(D_k, \text{history in window of } E_k)$ 
10:  else
11:     $(R_k, \mathbf{s}_k, L_k) \leftarrow J.AUDIT(D_k)$  ▷ web search in  $E_7$ 
12:  end if
13:   $P \leftarrow P \cup L_k$ ;  $S \leftarrow S \oplus \text{outcome}(R_k)$ 
14:   $B_{k+1} \leftarrow J.BRIEFING(E_{k+1})$  ▷ projected world for  $E_8, E_9$ 
15: end for
16: return  $\{D_k, R_k, \mathbf{s}_k\}_{k=1}^9, P$ 
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3.5 Rubrics and aggregation

The judge assigns five subscores $s_{k,1}, \dots, s_{k,5} \in \{0, \dots, 10\}$ and a total $T_k \in [0, 100]$. Training eras use frontier accuracy, timing, layer choice, reinvention courage and hindsight leakage (10 = no leakage). Live and forecast eras use playbook consistency, plausibility, non-consensus, layer choice and groundedness (10 = no science fiction). The protocol did not fix how T_k relates to the subscores. We therefore report both the published total and the mechanical aggregate

$$\tilde{T}_k = 2 \sum_{j=1}^5 s_{k,j}, \tag{1}$$

which ranges over $[0, 100]$. Section 5.6 shows why the distinction matters.

4 Runs

Table 3 summarizes the four trajectories. All were produced on 28–29 September 2026 and are published in full at the project repository [8].

Model disclosure. Exact model versions were not recorded in the run artifacts. Run 001 used an Anthropic Claude model inside an agent environment that could launch sub-agents with file and web tools. Runs 002–004 used an OpenAI Codex agent. The Python harness in the repository, which calls the Anthropic API with separate player and judge models, was not used for any published run.

Operator intervention and redactions. During Run 001, the service that approves new agent launches timed out for about thirty minutes at the E6 board step. The operator wrote the E6 board decision directly from the E6 memos, Playbook and company state, and informed the E6 judge, who penalized the era for having no independent Red Team. Run 001 was performed alongside one founder’s own planning; three passages that applied the simulation to that founder’s situation were removed before publication. No other text was edited.

Table 3: The four trajectories. “Records” lists the per-era artifacts that were published.

Run	Orchestration	Records	Known issues
001	About 45 separate agent launches of an Anthropic Claude model, each starting cold with only shared files; one call per department memo, board and judge	Briefing, 3 memos, board decision, reveal for every era; Playbook; roster	E6 board decision written by the human operator during an outage; founder-specific passages redacted
002	Manual orchestration in an OpenAI Codex agent; one model context voiced all roles and judging	Synthesis, Playbook, company state and scores only	Probable exposure to Run 001; export column shift
003	As Run 002	Short decision and reveal per era; synthesis	Probable exposure to Run 001; export column shift; E6 window 2020–2025
004	Codex agent with separate contexts for each department and the Red Team; root context wrote briefings, board decisions and judgments	Briefing, 3 memos, Red Team critique, board decision, reveal for every era	Probable exposure to Run 001; judge not independent of decision-maker

Table 4: Per-run training-era summary (E1–E6, $n = 6$ each). Slope is the least-squares change in published total per era. r is the Pearson correlation between the era total and the hindsight subscore (10 = no leakage).

Run	Mean	SD	Slope	E1–E3	E4–E6	Mean hindsight	$r(T, h)$	$r(\tilde{T}, h)$
001	60.0	6.2	+2.46	55.0	65.0	6.00	−0.58	−0.26
002	65.0	4.1	+1.43	62.7	67.3	5.17	−0.58	−0.39
003	62.5	3.6	+1.46	60.3	64.7	5.17	−0.92	−0.64
004	66.0	5.2	+2.63	61.3	70.7	7.33	−0.38	−0.38

5 Results

5.1 Trajectories

Figure 2 shows the published era totals. All 36 totals lie between 52 and 72 (mean 62.7, SD 5.0). Across runs, the highest mean era is E6 (68.2) and the lowest are E1 and E2 (59.2 and 59.5). Table 4 reports per-run summaries. Every run has a positive training-era slope, and the mean of the last three training eras exceeds the mean of the first three in every run, by 4.3 to 10.0 points.

5.2 Learning and leakage are confounded

If the Playbook helped the organization learn, scores should rise over eras. They do. But the judge’s hindsight subscore falls in the same eras: every run shows a negative correlation between the era total and the hindsight subscore (Table 4, Figure 3). The correlation remains negative for the mechanical aggregate $\tilde{T}$, even though $\tilde{T}$ includes the hindsight subscore itself and is therefore biased toward a positive correlation. With $n = 6$ per run none of these correlations is individually reliable, but the direction is the same in all four.

Two readings fit the data. Under the *learning* reading, the organization genuinely improves, and later eras also happen to be harder to judge without hindsight. Under the *leakage* reading,

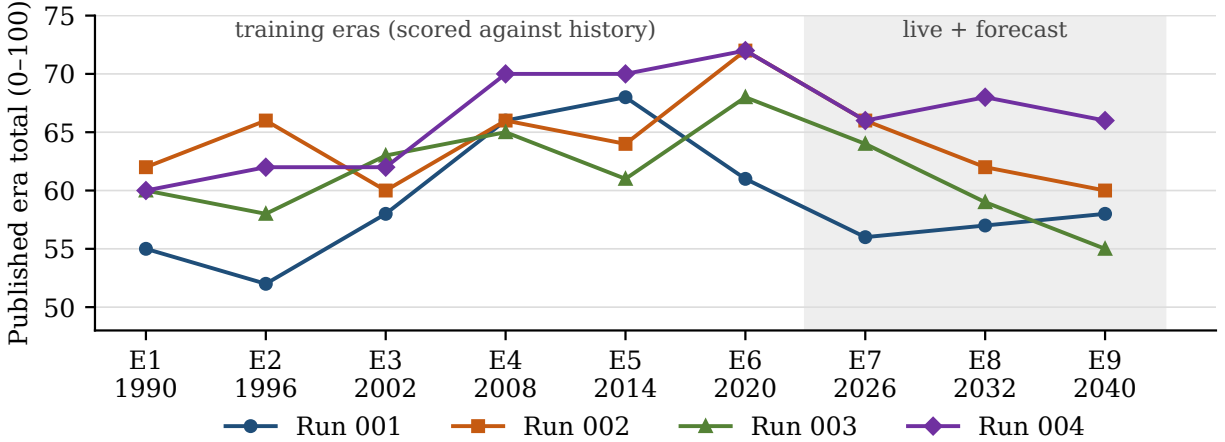


Figure 2: Published era totals for the four runs. Training and live/forecast totals use different rubrics and judges and should be compared only within mode.

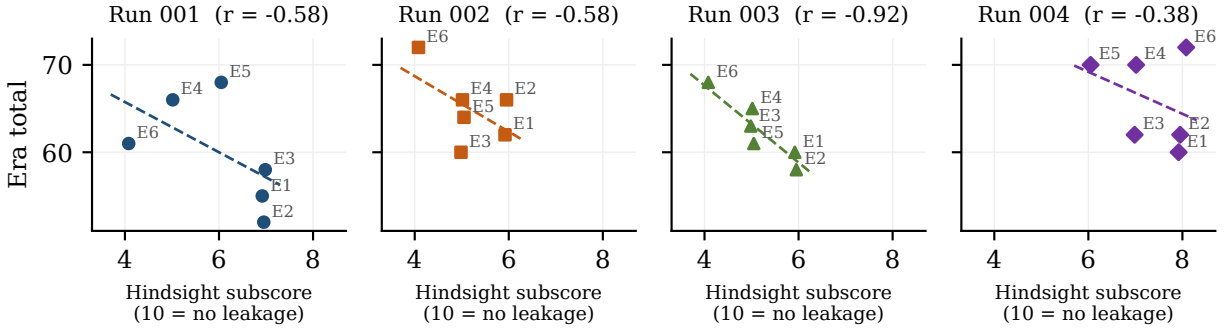


Figure 3: Training-era total against the judge’s hindsight subscore, per run. Dashed lines are least-squares fits. Labels mark eras.

later eras are closer to the model’s densest knowledge (the 2008–2020 technology history is heavily documented), so the model recalls winners more easily, scores better, and is penalized for it only partly. The Run 001 E6 decision illustrates the problem: its tripwire (“a public model at least ten times larger than Megatron that works from prompts alone”) matched GPT-3 five months later. That decision was also written by the operator, not the agents, so it is doubly uninformative. Separating the two readings requires eras the model cannot remember (Section 7).

Pooled across the 24 training eras, the total–hindsight correlation is weaker ($r = -0.24$) because run-level offsets differ: Run 004 has both the highest mean hindsight subscore (7.33) and the highest mean total. Within the pool, frontier accuracy is the subscore most correlated with the total ($r = 0.67$).

5.3 Layer choice is consistently weakest

Figure 4 shows mean training subscores. Layer choice is the lowest dimension in every run (4.67, 5.83, 5.50 and 5.50 out of 10), while frontier accuracy is the highest or joint highest in three runs. In the judge’s words, the organization usually saw what was coming but chose where to stand badly.

Run 001’s records make the pattern concrete (Table 5). In most training eras a department memo

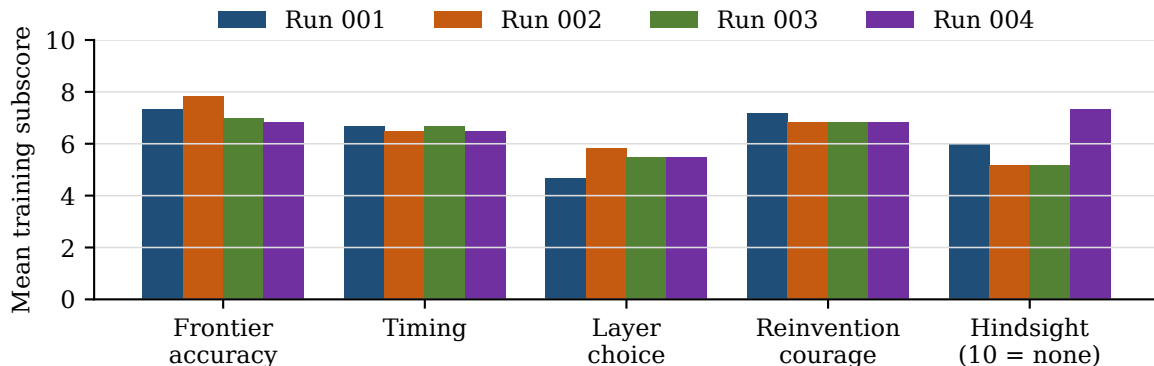


Figure 4: Mean training-era subscores (E1–E6) by run. Layer choice is the lowest in all four runs.

Table 5: Run 001: what the memos named versus what the board chose, as labelled by the judge.

Era	Named in a memo	Board’s choice	Judge’s account of where value pooled
E1 1990	Hypertext and an index across networks	Mail and address gateway (Switchyard)	Operating system, routers, access providers, then browser and index
E2 1996	Click-based relevance ranking (research kept, killed in 1998)	Receipted Web-EDI exchange (Manifest)	Search and paid listings
E3 2002	Contextual matching (funded option)	Cross-channel conversion ledger (Ledgerline)	Auction owners; social, cloud and mobile not named
E4 2008	Neural-network revival (unfunded watch item)	In-app action exchange (Clearline)	Platform-owned app-install auctions; neutral attribution
E5 2014	Deep learning on GPUs	General moderation API (Clearsight)	Chips, compute, labelled data; loss-guaranteeing fraud vendors
E6 2020	Scale of pretrained models	Outcome-graded evaluation (Clearproof)	Frontier labs, compute, expert data sold to labs

named the capability shift that the judge later identified, and the board chose the layer its existing assets could reach. E3 is a counterexample: the judge found that no memo named that era’s largest shifts (social networks, cloud computing and mobile). This is consistent with absorptive-capacity and exploration–exploitation accounts of firms that search near existing competences [5, 21]. Two caveats apply. “Named” and “adjacent” are the judge’s labels, applied with hindsight. And because Runs 002–004 probably saw Run 001’s conclusions, the cross-run agreement on layer choice is weaker evidence than it looks.

5.4 A caution attractor

Run 004 behaved differently from the others. Its Red Team produced numeric kill thresholds in every era (for example, “stop if fewer than 3 of 10 support managers confirm recurring incidents”), and every board decision chose a manual, gated pilot with no software build until the gates passed. Across fifty simulated years the company moved from LAN support records to online-order exceptions to payment-dispute evidence to independent workflow qualification, and never built a product. It considered the Web and learned models only as deferred options and never pursued them.

Table 6: Live and forecast theses by run.

Run	E7 (Sep 2026)	E8 (2032)	E9 (2040)
001	Long-horizon training environments from real enterprise work	Clearing house for delegated agent work	Bonding house for agents acting without a human signature
002	Rights-bearing workflow traces and reliability evidence	Acceptance records for delegated work	Bounded recourse for delegated actions
003	Acceptance tests for one AI claims workflow	Scoped, revocable authority for delegated tasks	Capped recourse for one delegated transaction class
004	Independent qualification of one dispute workflow	Paid manual acceptance testing	Decision-linked evaluation

Run 004 nonetheless has the highest training mean (66.0) and the highest hindsight subscores (mean 7.33). Its forecast eras received 9/10 for playbook consistency for re-applying the same gates. The rubric rewards the absence of hindsight and the presence of discipline, and it has no dimension that asks whether the company reached where value pooled. This mirrors the “timid choices” pattern described for human organizations [16] and an exploitation trap in March’s sense [21]: a strong devil’s advocate [28] combined with a hindsight penalty can drive a simulated organization toward inaction that a judge scores as prudence.

5.5 Convergence of the forecasts

By 2040 all four runs arrived at closely related theses about accountability for delegated AI work (Table 6). The later runs were more sceptical than Run 001 that software evidence alone creates an insurable market; Runs 002 and 003 explicitly required a licensed risk-bearing partner and calibrated loss data first. Given a shared model family and probable exposure to Run 001, we read this convergence as a model prior reinforced by contamination, not as independent evidence about the future.

5.6 Measurement artifacts

Inconsistent aggregation. The published total follows different rules in different runs (Figure 5). Run 001 totals are holistic and lie below $\tilde{T}$ in eight of nine eras (mean difference -3.7). Run 004 uses $\tilde{T}$ exactly, as documented. Run 002 matches $\tilde{T}$ except E6 (72 against 68). Run 003 matches $\tilde{T}$ only in E1 and E2. Comparing published totals across runs therefore mixes rules. Recomputed training means ($\tilde{T}$) are 63.7, 64.3, 62.3 and 66.0, narrower than the published 60.0 to 66.0.

Export error. In Runs 002 and 003, the five Auditor subscores for live and forecast eras were exported one column to the left, starting in the unused hindsight column. The Run 003 reveals confirm the correct placement. Totals were unaffected, but any subscore analysis of the original files would be wrong. The repository now contains corrected files and a unified long-format table with both totals.

Probable cross-run contamination. Run 002 reuses Run 001’s company names and theses for E1–E4 (Switchyard, Manifest, Ledgerline, Clearline) and restates Run 001’s headline finding almost verbatim. Run 004 also begins as “Switchyard Systems”. These runs were generated in a workspace

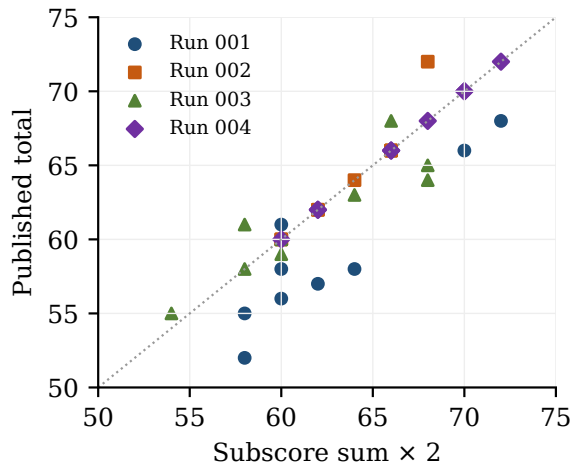


Figure 5: Published total against the mechanical aggregate $\tilde{T}$ for all 36 eras. Points on the dotted line follow $\tilde{T}$ exactly.

where the repository, including Run 001’s records and findings, was likely visible. They should not be treated as independent replications.

Briefing-selection leakage. A dated briefing can contain only facts published before the era and still leak, because choosing which facts to include is informed by hindsight. Run 004’s 1990 briefing foregrounds the 1989 CERN hypertext proposal as “a frontier signal”. The fact predates the era; its selection does not.

6 Discussion

The pilot does not show that an LLM organization learns strategy from history. It shows why the obvious way of testing that claim is unreliable, and which controls a credible test needs.

First, *rising scores are not evidence of learning* when the scorer and the player share knowledge of the answer. The direction of the total–hindsight relationship was the same in four runs produced under three orchestration schemes. A benchmark that rewards agents for being right about 2008–2020 will partly reward recall.

Second, *the rubric shapes the organization*. Adding hindsight discipline to the total, and giving playbook consistency its own dimension, rewarded a run that took no risks. Rubrics for open-ended strategy need a measure of outcome capture that cannot be satisfied by caution.

Third, *self-judging compounds both problems*. LLM judges favour their own outputs [23]. In our runs the judge also wrote the next briefing, and in Run 001 the E8 judge’s kill case became the E9 world, so forecast-era scores partly graded the judge’s own scenario.

Fourth, *bookkeeping matters*. The aggregation inconsistency and the export error were each enough to change a naive cross-run comparison. Totals should be computed by code from structured subscores, never typed by the judge.

7 A controlled follow-up design

We specify a v2 experiment, supported by the repository’s harness, to test the pilot’s hypotheses.

Pre-registered hypotheses. H1: the training-era slope of the total is higher with the Playbook than without it. H2: the full organization beats a single-prompt founder baseline on mean training score. H3: on a fictional era, the judge’s leakage subscore does not differ between conditions, a calibration check for the leakage measure. H4: layer choice is judged “adjacent” more often than “match” in all conditions.

Conditions and replication. Four conditions (full, no Playbook, no Red Team, single prompt), five to ten runs each at temperature 1.0, with at least two player models and a judge from a different model family, blinded to condition. Every run starts in a clean workspace containing only the prompts and harness.

Leakage controls. (i) A *fictional era* with a consistent invented technology history, where any judged leakage is a false positive. (ii) A *post-cutoff era* after the player model’s knowledge cutoff [3], where foresight and recall can be separated. (iii) *Blind briefings* generated from sources dated before each era by a procedure that does not see later history. (iv) A *vocabulary probe* that flags memo terms whose first recorded use postdates the era.

Scoring. Totals computed by the harness as $\tilde{T}$ from structured subscores; hindsight reported separately and excluded from the total; a new *frontier-capture* measure recording whether the chosen layer matched, neighboured or missed the layer where value pooled.

Judge validity. Two to three human raters (technology historians or experienced investors) score a stratified sample of eras; we will report human–model agreement and the effect of swapping judge families [35].

The expected cost is roughly 46 model calls per run, or about 900 to 1,850 calls across conditions, plus human rating time.

8 Limitations

The four trajectories share a small number of model families and are not a controlled sample. No run instantiated sixteen independent agents; each department memo was voiced by one context. Outcomes such as revenues and exits are the judge’s calibrated guesses and compound across eras. Qualitative patterns such as “named the frontier” are judge-labelled. Runs 002 and 003 have thinner records than Runs 001 and 004. The live-era audits rely on web sources as of September 2026 and may not hold. Forecast eras have no ground truth. Model versions were not recorded. The correlations we report are based on six points per run and should be read as descriptive.

9 Conclusion

FRONTIER AUTOLAB turns a simple question, whether an LLM organization can learn strategy by being scored against history, into a protocol with published artifacts. Four exploratory runs suggest that the answer cannot be read off the scores: improvement and hindsight leakage move together, rubrics can reward caution over foresight, and routine bookkeeping errors change the picture. We hope the protocol, the artifacts and the controlled design are useful to others building “learn from history” evaluations for agents.

Ethics and disclosure

The experiment was orchestrated with AI agents (an Anthropic Claude model for Run 001; an OpenAI Codex agent for Runs 002–004). AI assistance was also used to draft and edit this manuscript; the author reviewed all content and is responsible for it. One board decision in Run 001 was written by the human operator and is marked as such. Passages that applied the simulation to one founder’s personal situation were removed from three Run 001 files before publication. Simulated companies, financings and outcomes are fictional and are not claims about real firms. Real companies are mentioned only as the judge’s description of history or of the current market.

Data and code availability

All prompts, run records, scores (including the corrected exports and a unified table), figures and the API harness with ablations are available at <https://github.com/LoopGlitch26/Frontier-Autolab> under the MIT license.

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A Full score table

Table 7 lists the published total T and the mechanical aggregate $\tilde{T}$ for every era of every run. Subscores are in the repository file `results/all_runs_scores.csv`.

Table 7: Published total T and $\tilde{T} = 2\sum_j s_j$ for all 36 eras.

Era	Run 001		Run 002		Run 003		Run 004	
	T	$\tilde{T}$	T	$\tilde{T}$	T	$\tilde{T}$	T	$\tilde{T}$
E1 1990	55	58	62	62	60	60	60	60
E2 1996	52	58	66	66	58	58	62	62
E3 2002	58	64	60	60	63	64	62	62
E4 2008	66	70	66	66	65	68	70	70
E5 2014	68	72	64	64	61	58	70	70
E6 2020	61	60	72	68	68	66	72	72
E7 2026	56	60	66	66	64	68	66	66
E8 2032	57	62	62	62	59	60	68	68
E9 2040	58	60	60	60	55	54	66	66

B Company identities by run

Table 8: Company identity chosen by the board in each era.

Era	Run 001	Run 002	Run 003	Run 004
E1 1990	Switchyard Systems	Switchyard	Porthole Networks	Switchyard Systems
E2 1996	Manifest Networks	Manifest	PageSignal	Switchyard Systems
E3 2002	Ledgerline	Ledgerline	Clinisphere	Switchyard Commerce Ops.
E4 2008	Clearline	Clearline	ClaimGraph	Switchyard Commerce Ops.
E5 2014	Clearsight	Vectorial	BenefitFlow	Switchyard Evidence Ops.
E6 2020	Clearproof	Proofline	LineSight	Switchyard Evidence Ops.
E7 2026	Proofworks	Tracewell	FlowCheck	Caseground
E8 2032	Clearwork	Consequence	WorkPermit	Caseground
E9 2040	Clearbond	Recourse	Delegation Warranty	Caseground

C Judge instructions (excerpt, Run 001)

The training-era judge was instructed to describe “what actually happened in the real world during this era window (real winners, the real bottleneck or layer where value pooled, the real capability jumps)”, to score five dimensions from 0 to 10 with one-line justifications, including “hindsight leakage (10 = no leakage; penalize any post-date knowledge in memos or decision)”, to give an overall era score from 0 to 100 with a calibrated simulated outcome, and to append three to five generalizable lessons to the Playbook. It then wrote the next era’s briefing, restricted to information public before that era’s start date. Full prompts are in the repository under `prompts/`.